

Beyond a VPC: egress, peering, hybrid, edge

Day 4 - Morning

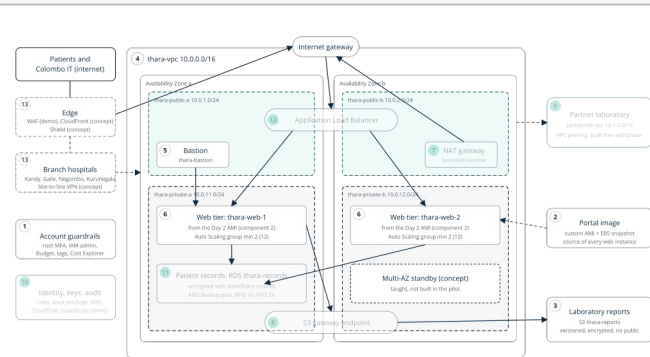
Running scenario: Thara Hospitals patient portal pilot

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The web tier has no way out. Today it gets three paths, each with a price.

- Getting out without being reachable - 30 min
- Talking to other networks - 35 min
- Names and the edge - 30 min
- Estimating the pilot - 15 min



13 Edge and branches

BUILT EARLIER

1

2

(2)

4

5

6

CONCEPT 1



Getting out without being reachable

Patch without a public address. Reach the reports without the internet.

30 min

The NAT gateway

- A managed service in a **public** subnet, holding an Elastic IP
- Private subnets send their default route to it
- It translates connections that start inside. Nothing outside can start one
- It lives in one zone, and bills **by the hour and by the gigabyte**
- A NAT instance is the cheaper alternative, and yours to patch

YOU ALREADY KNOW THIS

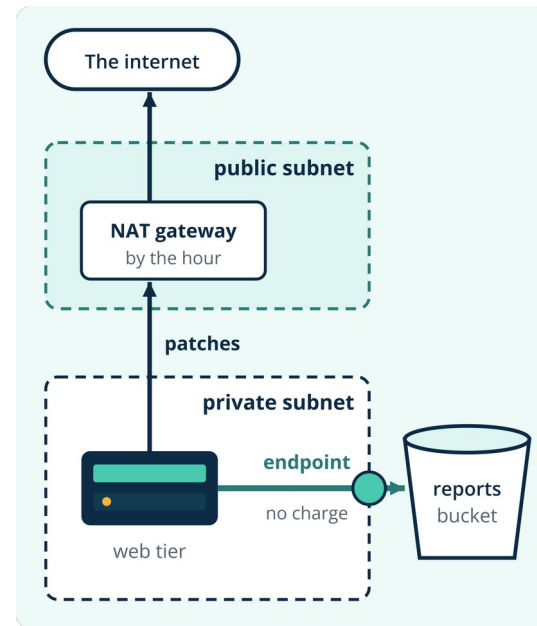
PAT on the edge router versus a private peering to a partner.

VPC endpoints

- S3 is a public service: a private subnet cannot reach it either
- A **gateway endpoint** for S3 is a route to a prefix list. It is free
- Interface endpoints exist for most other services, and are charged



The reports bucket must never be reached over the internet.



Patches by one path, reports by the other.

Why Thara uses both



Reports, through the endpoint

Health data, read all day, from one known service. A free path that never touches the internet.

Patches, through NAT

From the internet in general, now and then. A paid path, opened for an hour and deleted.

Left running for a month, about USD 41 before any traffic.

A NAT gateway is outbound only. It does not make `thara-web-1` reachable.

CONCEPT 2



Talking to other networks

The partner laboratory, the four branches, and a fifth site.

35 min

VPC peering

- **One-to-one:** a peering joins exactly two VPCs
- **Non-transitive:** A to B and B to C does not give A to C
- A route on **both sides:** the reply needs its own
- No overlapping ranges: the partner is 10.1.0.0/16

YOU ALREADY KNOW THIS

Static routes between two branch routers versus an MPLS hub.

COMMON MISTAKE

Peering is transitive.

It is not. Each pair that must talk needs its own peering, or a hub.

Transit Gateway: the hub

- Five networks, fully peered, need ten peerings
- A Transit Gateway is a regional hub: each network attaches once
- It fixes non-transitivity at scale, and it is charged
- It answers a question of scale. It is not where a pilot starts



THARA HOSPITALS

The partner laboratory is a peering. A fifth site is the Transit Gateway question.

Hybrid: VPN or Direct Connect

0.05

USD per VPN connection-hour
about, at the time of writing

1.25

Gbps for each VPN tunnel
and never above the site's own link

Weeks

to provision Direct Connect
a dedicated line through a carrier

Point	Site-to-Site VPN	Direct Connect
Path	Tunnels over the internet	A dedicated link through a carrier
Bandwidth	Up to about 1.25 Gbps a tunnel	Ports of 1 Gbps and upwards
Latency	Varies with the internet	Steady
Cost	Low	Expensive
Setup time	Hours	Weeks
At the site	A customer gateway device	A carrier circuit



IT

HEAD OF IT

Connect the branches to the pilot later, and do it without one new leased line. They reach us over the internet links they already have.

The Head of IT, on Kandy, Galle, Negombo and Kurunegala.

CONCEPT 3



Names and the edge

One name that survives an instance failure.

30 min

Route 53: names

- A hosted zone holds the records for one domain
- **A**: a name to an address. **CNAME**: a name to another name
- **Alias**: a name to an AWS resource, followed as its addresses change
- A **private hosted zone** answers only inside its VPCs

The demonstration creates a private zone for thara.internal.

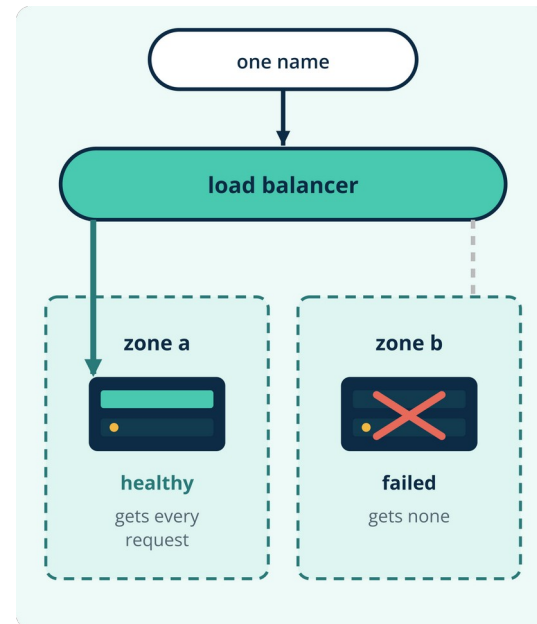
The Application Load Balancer

- Spreads requests across targets in more than one zone
- Asks each target a **health check**, and skips any that fails
- The first concrete high-availability mechanism

COMMON MISTAKE

The load balancer is a server.

It is a managed, multi-AZ service with its own DNS name.



The name stays. The servers behind it change.

At the edge



CACHE

CloudFront

Keeps copies of content close to users, so the web tier is asked less often.

ABSORB

Shield

Takes denial-of-service floods at the edge, before they reach your network.

FILTER

WAF

A rule engine that inspects web requests and blocks those that match.

Demonstrated on Day 7.

Component 13. Ideas today, in front of the portal in a later phase.

CONCEPT 4



Estimating the pilot

A calculator estimate inside USD 150.

15 min

The Pricing Calculator

- Build the Thara pilot line by line: region, service, usage
- A month is 730 hours
- The largest lines are the load balancer and the NAT gateway, if left running
- A **bounded window** is the saving: 40 hours is about one eighteenth of 730



THARA HOSPITALS

The capstone brief requires a calculator estimate inside USD 150.

Demonstration: Peering, names, and the calculator

1. Peer two VPCs. Connectivity fails until **both** route tables are updated
2. Create a private hosted zone for thara.internal with an A record for the web instance. Resolve it from the bastion
3. Build the pilot in the Pricing Calculator. The load balancer at 730 hours, then at 40

Watch the route tables: this afternoon they are yours.

Activity: The capstone brief and your group, 25 minutes



1. Walk through the capstone brief together
2. Form groups of three. Nominate the host account
3. Assign tier responsibilities: network, security, data
4. Write down three questions about member access, for the Day 5 lab

Common mistakes



COMMON MISTAKE

Peering is transitive.

It is not. A peering joins two networks and forwards for no third.

COMMON MISTAKE

A NAT gateway makes instances reachable from the internet.

It is outbound only.

COMMON MISTAKE

The load balancer is a server.

It is a managed, multi-AZ service with its own DNS name.

Before the next session



READ

Shields, AWS Security

Chapters 5 and 6: the VPC, and network access beyond the VPC.

SELF-CHECK

CLF-C02 bank

Domain 3 networking set C, 10 items.

PRACTICE

PQ4

Hybrid connectivity for the branches, with cost and latency reasoning.

Evidence Pack 4 follows this afternoon's lab.

Questions & Discussion



Next: this afternoon you build components 7, 8 and 9, and take two of them down.



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